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Program: Btech CSE B16

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HDOC Day-26

1. Algorithm

1. Start the program.
 2. Declare integer variable `choice`.
 3. Display the menu:
 - Full Pyramid
 - Inverted Pyramid
 - Diamond Pattern
 - Palindromic Number Pyramid
 - Repeated-Number Pyramid
 - Exit
 4. Accept `choice` from the user.
 5. Use `switch(choice)`.
 6. For choices 1-5, call the corresponding parameterless `void` function.
 7. For choice 6, exit the program.
 8. For any other choice, display "Invalid choice".
 9. Repeat the menu until the user chooses 6.
 10. Stop.
-

2. Pattern 1 — Full Pyramid

The first option requires `n` rows, with the number of asterisks increasing as 1, 3, 5,

Algorithm

1. Input `n`.
2. Run outer loop `i = 1` to `n`.

3. Print $n-i$ leading spaces.
4. Print $2*i-1$ asterisks.
5. Move to the next line.
6. Repeat until all rows are printed.

For $n = 4$

```

      *
     * * *
    * * * * *
   * * * * * * *
```

3. Pattern 2 — Inverted Pyramid

The second option starts with $2n-1$ asterisks and decreases toward one.

Algorithm

1. Input n .
2. Run outer loop from $i = n$ down to 1.
3. Print $n-i$ leading spaces.
4. Print $2*i-1$ asterisks.
5. Move to next line.
6. Continue until one asterisk remains.

For $n = 4$

```

* * * * *
 * * * *
  * * *
   *

```

4. Pattern 3 — Diamond Pattern

For input n , the middle row contains $2n-1$ asterisks and the complete diamond contains $2n-1$ rows.

Algorithm

1. Input n .
2. Print the upper half:
 - o Loop $i = 1$ to n .
 - o Print $n-i$ spaces.

- Print $2*i-1$ asterisks.
- 3. Print the lower half:
 - Loop $i = n-1$ down to 1.
 - Print $n-i$ spaces.
 - Print $2*i-1$ asterisks.
- 4. Return to main menu.

For $n = 4$

```

      *
    * * *
  * * * * *
* * * * * * *
  * * * * *
    * * *
      *

```

5. Pattern 4 — Palindromic Number Pyramid

Each row contains natural numbers increasing from 1 to the row number and then decreasing back to 1.

Algorithm

1. Input n .
2. Run outer loop $i = 1$ to n .
3. Print the required leading spaces.
4. Print numbers from 1 to i .
5. Print numbers from $i-1$ down to 1.
6. Move to next line.
7. Repeat for all rows.

For $n = 4$

```

      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1

```

6. Pattern 5 — Repeated-Number Pyramid

For each row, the row number is printed $2i-1$ times. For example, row 4 contains seven 4s.

Algorithm

1. Input n .
2. Run outer loop $i = 1$ to n .
3. Print $n-i$ leading spaces.
4. Print the value of i , $2*i-1$ times.
5. Move to next line.
6. Repeat until all rows are printed.

For $n = 4$

```
    1
  2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
```

7. Complete C Program

```
#include <stdio.h>

void fullPyramid();
void invertedPyramid();
void diamond();
void palindromePyramid();
void repeatedNumberPyramid();

int main()
{
    int choice;

    do
    {
        printf("\n===== PYRAMID / DIAMOND MENU =====\n");
        printf("1. Full Pyramid\n");
        printf("2. Inverted Pyramid\n");
        printf("3. Diamond Pattern\n");
        printf("4. Palindromic Number Pyramid\n");
        printf("5. Repeated-Number Pyramid\n");
        printf("6. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                fullPyramid();
                break;

            case 2:
                invertedPyramid();
```

```

        break;

    case 3:
        diamond();
        break;

    case 4:
        palindromePyramid();
        break;

    case 5:
        repeatedNumberPyramid();
        break;

    case 6:
        printf("Exiting the program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to 6.\n");
}

} while(choice != 6);

return 0;
}

```

```

/* 1. Full Pyramid */
void fullPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++)
            printf("* ");

        printf("\n");
    }
}

```

```

/* 2. Inverted Pyramid */
void invertedPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

```

```

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++)
            printf("* ");

        printf("\n");
    }
}

/* 3. Diamond Pattern */
void diamond()
{
    int n, i, j;

    printf("Enter number of rows for upper half: ");
    scanf("%d", &n);

    /* Upper half */
    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++)
            printf("* ");

        printf("\n");
    }

    /* Lower half */
    for(i = n - 1; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++)
            printf("* ");

        printf("\n");
    }
}

/* 4. Palindromic Number Pyramid */
void palindromePyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {

```

```

        for(j = 1; j <= n - i; j++)
            printf(" ");

        /* Increasing numbers */
        for(j = 1; j <= i; j++)
            printf("%d ", j);

        /* Decreasing numbers */
        for(j = i - 1; j >= 1; j--)
            printf("%d ", j);

        printf("\n");
    }
}

/* 5. Repeated-Number Pyramid */
void repeatedNumberPyramid()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= 2 * i - 1; j++)
            printf("%d ", i);

        printf("\n");
    }
}

```

8. Test-Case Table

TC Choice	n	Expected Output	Actual Output	Result
1	1	4 Full pyramid of 4 rows	Same as expected	Pass
2	2	4 Inverted pyramid of 4 rows	Same as expected	Pass
3	3	4 Diamond containing 7 rows	Same as expected	Pass
4	4	4 Palindromic number pyramid	Same as expected	Pass
5	5	4 Repeated-number pyramid	Same as expected	Pass

TC Choice	n	Expected Output	Actual Output	Result
6	6	— Program terminates	Program terminated	Pass
7	8	— Invalid choice message	Invalid choice displayed	Pass

9. Test Execution

Test Case 1-

```

(kali㉿kali)-[~/Desktop]
└─$ cc day26.c

(kali㉿kali)-[~/Desktop]
└─$ ./a.out

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 1
Enter number of rows: 4
  *
 * *
* * *
* * * *
* * * * *

```

Test Case 2-

```

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 2
Enter number of rows: 4
* * * * *
 * * * *
  * * *
   * *
    *

```

Test Case 3-

```

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 3
Enter number of rows for upper half: 4
  *
 * *
* * *
* * * *
* * * * *
  * * *
   * *
    *

```

Test Case 4-

```

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 4
Enter number of rows: 4
  1
 1 2 1
1 2 3 2 1
1 2 3 4 3 2 1

```

Test Case 5-

```

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 5
Enter number of rows: 4
  1
 2 2 2
 3 3 3 3
 4 4 4 4 4 4

```

Test Case 6-

```

===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 6
Exiting the program...

```

Test Case 7-

```
===== PYRAMID / DIAMOND MENU =====
1. Full Pyramid
2. Inverted Pyramid
3. Diamond Pattern
4. Palindromic Number Pyramid
5. Repeated-Number Pyramid
6. Exit
Enter your choice: 8
Invalid choice! Please enter 1 to 6.
```

Conclusion

The program implements all **five patterns** specified in HDOC Day-26 using **switch-case**, **nested loops**, **no arrays**, and **separate parameterless void functions**, as required by the assignment.